Section 16.8: Stokes' Theorem

Vector Calculus

MTH 310
Calculus III

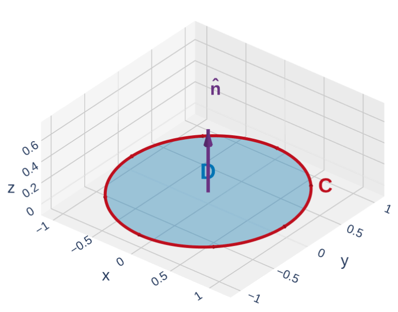
The Big Picture: Green's Theorem in 3D

In Section 16.4, **Green's Theorem** connected a line integral around a **flat** closed curve to a double integral over the flat region it bounds:

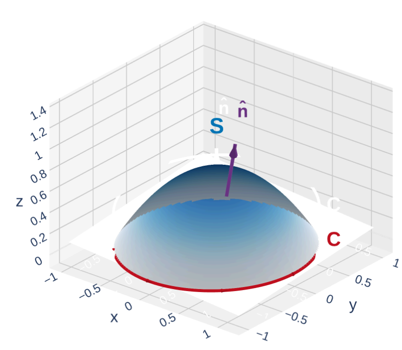
$$\oint_C P\,dx + Q\,dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dA$$

Stokes' Theorem generalizes this: replace the flat region by *any* oriented surface S in 3D, and replace C by the (possibly non-planar) boundary curve of S.

Green's (Section 16.4): flat disk D in the *xy*-plane



Stokes' (Section 16.8): curved surface S in 3D



Interactive version

Think of the flat disk being **lifted and bent** into a curved surface like a soap film stretched across a bent wire. The curve C is the wire; S is the film.

> Green's Theorem is the flat special case of Stokes' Theorem. Stokes' works for any oriented surface in $\mathbb{R}^3$ whose boundary is a piecewise-smooth closed curve.

Stokes' Theorem

Theorem: Stokes' Theorem

Let S be an oriented, piecewise-smooth surface bounded by a simple, closed, piecewise-smooth boundary curve C with **positive orientation**. Let $\vec{F}$ be a vector field whose components have continuous partial derivatives on an open region containing S. Then

$$\boxed{\int_C \vec{F}\cdot d\vec{r} \;=\; \iint_S \operatorname{curl}\vec{F}\cdot d\vec{S}.}$$

In words: the **circulation** of $\vec{F}$ around the boundary curve equals the **flux of curl** $\vec{F}$ through the surface.

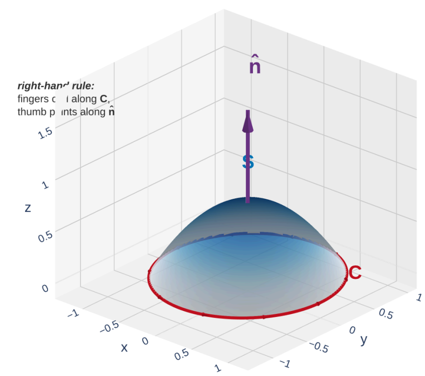
Reading each piece:

- $\displaystyle\int_C \vec{F}\cdot d\vec{r}$ — line integral around the boundary (work / circulation)
- $\operatorname{curl}\vec{F} = \nabla\times\vec{F}$ — measures local rotation of the field
- $\displaystyle\iint_S \operatorname{curl}\vec{F}\cdot d\vec{S}$ — total curl flux through S

Stokes': *line integral around the edge* $=$ *surface integral of the curl over the interior*. Exactly the same template as Green's, one dimension lifted.

Orientation: The Right-Hand Rule

For Stokes' Theorem to work, the orientations of C and S must be **compatible**.



Interactive version

Right-hand rule: curl the fingers of your **right hand** in the direction of travel along C; your **thumb** points in the direction of the surface normal $\vec{n}$.

Equivalently: walk along C in the positive direction with your head pointing along $\vec{n}$ — the surface is on your **left**.

Shortcut for graphs. If S is oriented **upward** (normal points up), then C is traversed **counterclockwise** when viewed from above.

The right-hand rule ties C's direction to S's normal. Getting it backward flips the sign of the answer — always double-check before computing.

Green's Theorem Is a Special Case of Stokes'

Take S = a flat region D in the xy-plane, oriented upward ($\vec{n} = \vec{k}$), with $\vec{F} = \langle P, Q, 0 \rangle$ (no z-component, components independent of z).

Curl: Since P, Q don't depend on z,

$$\operatorname{curl} \vec{F} = \left\langle 0,\ 0,\ \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right\rangle.$$

Flux through the flat disk:

$$\iint_S \operatorname{curl} \vec{F} \cdot d\vec{S} = \iint_D \langle 0,\ 0,\ Q_x - P_y \rangle \cdot \langle 0,\ 0,\ 1 \rangle\, dA = \iint_D (Q_x - P_y)\, dA.$$

And Stokes' gives

$$\oint_C P\, dx + Q\, dy = \iint_D (Q_x - P_y)\, dA,$$

which is exactly **Green's Theorem**.

Green's (flat xy-plane) is Stokes' restricted to a horizontal disk. Stokes' opens the same trick up to any oriented surface whose boundary is a simple closed curve.

Quick Check: Identify the Pieces

Let S be the part of the paraboloid $z = 9 - x^2 - y^2$ above the xy-plane, oriented upward. What is the boundary curve C?

Set $z = 0$ in the equation of the surface. What shape do you get, and what is its orientation?

Example 1: Line Integral via Stokes' Theorem

Example 1

Let $\vec{F}(x, y, z) = \langle 2y\cos z,\ e^x \sin z,\ x\,e^y \rangle$, and let C be the circle $x^2 + y^2 = 9$ in the plane $z = 0$, oriented counterclockwise. Evaluate $\displaystyle\int_C \vec{F} \cdot d\vec{r}$.

Direct parametrization of C gives a messy $e^{3\sin t}$ term. Use Stokes' with the simplest surface bounded by C — the flat disk in the xy-plane.

The Power of Stokes': Choose the Easier Side

Stokes' gives us a **choice**: compute the line integral side or the surface integral side — whichever is simpler.

Bonus freedom: the same curve C bounds *many* surfaces. Pick the surface that makes $\iint_S \operatorname{curl} \vec{F} \cdot d\vec{S}$ easiest.

Stokes' is a **computational tool**: convert a hard integral into an easy one. The surface integral depends only on the curl and the boundary C, not on which surface filling you pick.

Example 2: Line Integral via Stokes'

Example 2

Evaluate $\int_C \vec{F} \cdot d\vec{r}$ where $\vec{F} = \langle 1,\ x + yz,\ xy - \sqrt{z} \rangle$ and C is the triangular boundary of the part of the plane $3x + 2y + z = 1$ in the first octant, oriented counterclockwise from above.

Computing the line integral directly means three separate segment integrals — painful. Use Stokes' with S = the triangular piece of the plane. Compute curl first.

Quick Check: Compute the Curl

Let $\vec{F} = \langle yz,\ xz,\ xy \rangle$. Compute $\operatorname{curl} \vec{F}$.

Formula: $\operatorname{curl} \vec{F} = \langle R_y - Q_z,\ P_z - R_x,\ Q_x - P_y \rangle$ where $\vec{F} = \langle P, Q, R \rangle$.

Stokes' and Conservative Fields (3D)

Stokes' finally **proves** the conservative-field theorem we stated without proof in Section 16.5.

Recall (16.5): if $\vec{F} = \nabla f$, then $\operatorname{curl}(\nabla f) = \vec{0}$.

Converse (stated, not proved before): if $\operatorname{curl} \vec{F} = \vec{0}$ on a *simply connected* region, then $\vec{F}$ is conservative.

Proof via Stokes'. If $\operatorname{curl} \vec{F} = \vec{0}$ throughout the region and C is any closed curve bounding a surface S in that region:

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_S \operatorname{curl} \vec{F} \cdot d\vec{S} = \iint_S \vec{0} \cdot d\vec{S} = 0.$$

Every closed-loop integral is zero, so $\vec{F}$ is path-independent — i.e., conservative.

The full 3D equivalence on a simply connected domain:

$$\vec{F} \text{ conservative} \iff \operatorname{curl} \vec{F} = \vec{0} \iff \oint_C \vec{F} \cdot d\vec{r} = 0 \text{ for all closed } C.$$

“Simply connected” in 3D means every closed curve bounds some surface inside the region. That's what lets Stokes' close the loop back to a conservative potential.

Example 3: Choosing a Better Surface

Example 3

Evaluate $\oint_C \vec{F} \cdot d\vec{r}$ where $\vec{F} = \langle -y^2,\ x,\ z^2 \rangle$ and C is the intersection of the plane $y + z = 2$ with the cylinder $x^2 + y^2 = 1$, oriented counterclockwise from above.

C is a tilted ellipse — parametrizing it directly is awkward. Use Stokes' with S = the flat piece of the plane inside the cylinder.

Quick Check: Flat-Disk Shortcut

Let $\vec{F} = \langle z, x, y \rangle$ and C = unit circle in the xy-plane, counterclockwise. Evaluate $\oint_C \vec{F} \cdot d\vec{r}$.

Use Stokes' with S = unit disk in the xy-plane. Only the $\vec{k}$-component of the curl survives the dot with $\vec{n} = \vec{k}$.

Example 4: Surface Integral via Line Integral

Example 4

Evaluate $\iint_S \operatorname{curl} \vec{F} \cdot d\vec{S}$ where $\vec{F} = \langle xz,\ yz,\ xy \rangle$ and S is the part of the sphere $x^2 + y^2 + z^2 = 4$ inside the cylinder $x^2 + y^2 = 1$ with $z \geq 0$, oriented upward.

The surface is a complicated spherical cap. But its boundary is a simple circle. Use Stokes' in the reverse direction: compute the line integral instead.

Physical Interpretation: Circulation and Curl

Curl $=$ microscopic circulation density. Stokes' integrates that local density over S to produce the macroscopic circulation around C.

Homework

Section 16.8: 1, 3, 5, 7, 9, 13, 17, 19